(T1), piezo buzzer (PZ1) and a few other components. In this design, PIC18F45K22 is used with LCD1 display and buzzer PZ1. It is clocked using an 8MHz crystal.

A compatible RX2A UHF receiver module (Fig. 8) is used at the monitoring station. This module has a sensitivity of -108dBm and operates with power supply from 2.7V to 16V. Typical current consumption is 11.6mA. Use of the module requires connection of an external antenna. Receiver data pin is connected to pin 26 or UART input (RX) of the MCU. Table II gives the pin configuration of RX2A.

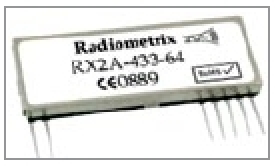

Fig. 8: RX2A UHF receiver module

Fig. 9: Displaying an alarm condition from station 3

TABLE I
PIN CONFIGURATION OF TX2H

TX2H pin	Name	Function
1	RF gnd	RF ground
2	RF out	50-ohm RF output to antenna
3	V_{CC}	+5V DC supply input
4	GND	DC supply ground
5	TXD	Transmit data input

Software

The software of the system has been developed using mikroC Pro for PIC language (www.mikroe.com). mikroC is a standard C-compatible language developed specifically for PIC MCUs. It has a rich library of functions, including RS-232, SPI, I2C, LCD, GLCD, TFT, SD card and PWM.

Detection station. At the beginning of DTransmitter.c program, threshold is set to 2500mV, so that the system is activated when LPG concentration level is approximately above 800ppm (Fig. 4). Then, Port B is configured as digital input and RA1 is configured as analogue input.

Port B internal pullups are enabled and UART is initialised to operate at 9600 baud rate. The rest of the program executes in an endless loop.

TABLE II
PIN CONFIGURATION OF RX2A

TX2H pin	Name	Function
1	RF in	50-ohm input from antenna
2	RF gnd	RF ground
3	RSSI	Received signal strength
4	GND	DC supply ground
5	V_{CC}	DC power supply
6	AF out	Analogue output from FM demodulator
7	RXD	Received data input

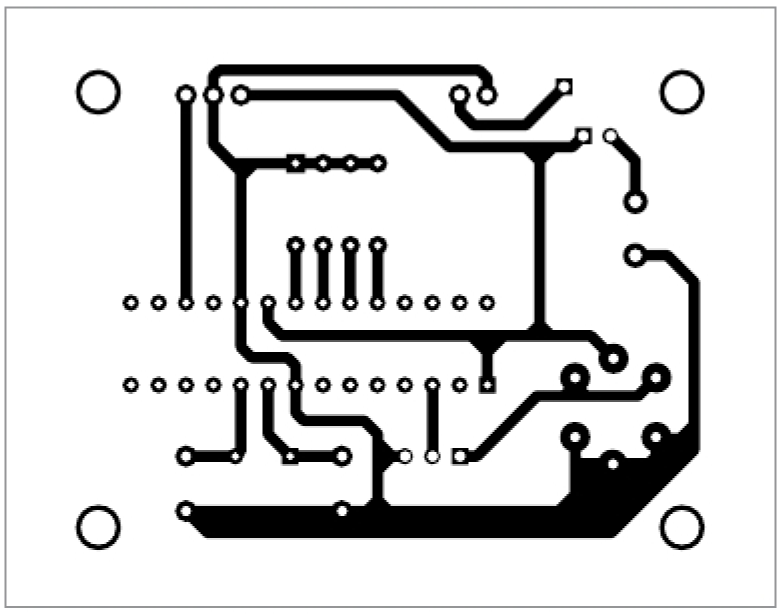

Fig. 10: Actual-size PCB layout of detection circuit

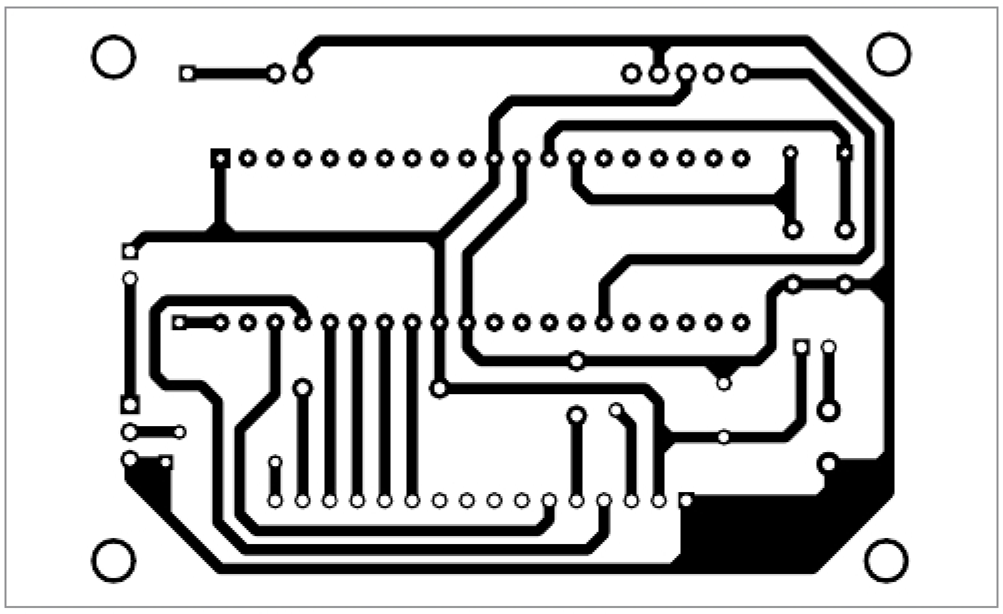

Fig. 12: Actual-size PCB layout of monitoring circuit

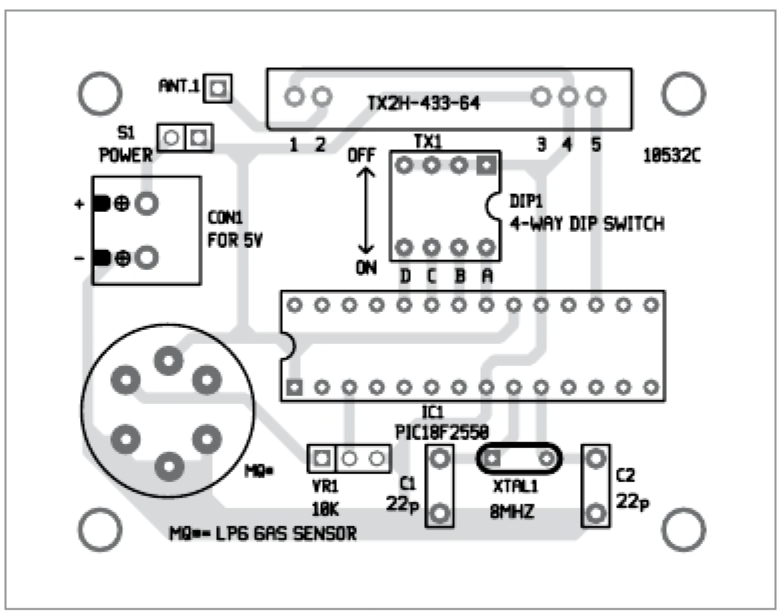

Fig. 11: Component layout for the PCB layout of detection circuit

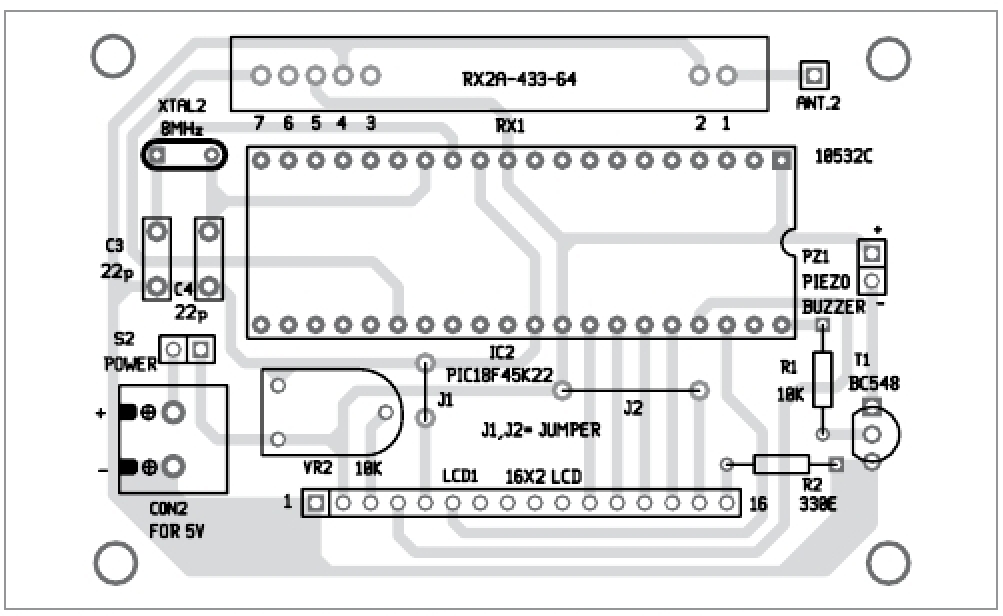

Fig. 13: Components layout for the PCB layout of monitoring circuit